AI ACADEMY · LEVEL 2 · GRADE 7 · WEEK 3 · STUDENT LESSON

Branching Logic

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

THE CONNECTED WEEK

Read the rule. Trace the first true branch. Test equality.

Write an ordered duration sorter, expose overwritten labels and compare clearly labelled category contracts.

Keep one habit: Specify categories → trace checked conditions → test the exact boundary and retain failures.

MAIN LESSON · SECTION 1

Which condition is checked next—and which is skipped?

> Trace ordered branches and test the equality that changes a category.

Start with the first true branch

The first true branch decides the path in an if/elif/else chain. Write the intended categories first, then test an exact boundary and inspect which later conditions were skipped.

Recall inclusive versus strict comparison

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Does minutes <= 12 accept exactly 12? How does < differ?

Your learning goals

- I can trace the selected **branch** in an if elif else chain.
- I can use **indentation** to identify which statements belong to a **branch**.
- I can test values at and around a decision **boundary**.

▼ Earlier wording, alternatives and picture transcripts

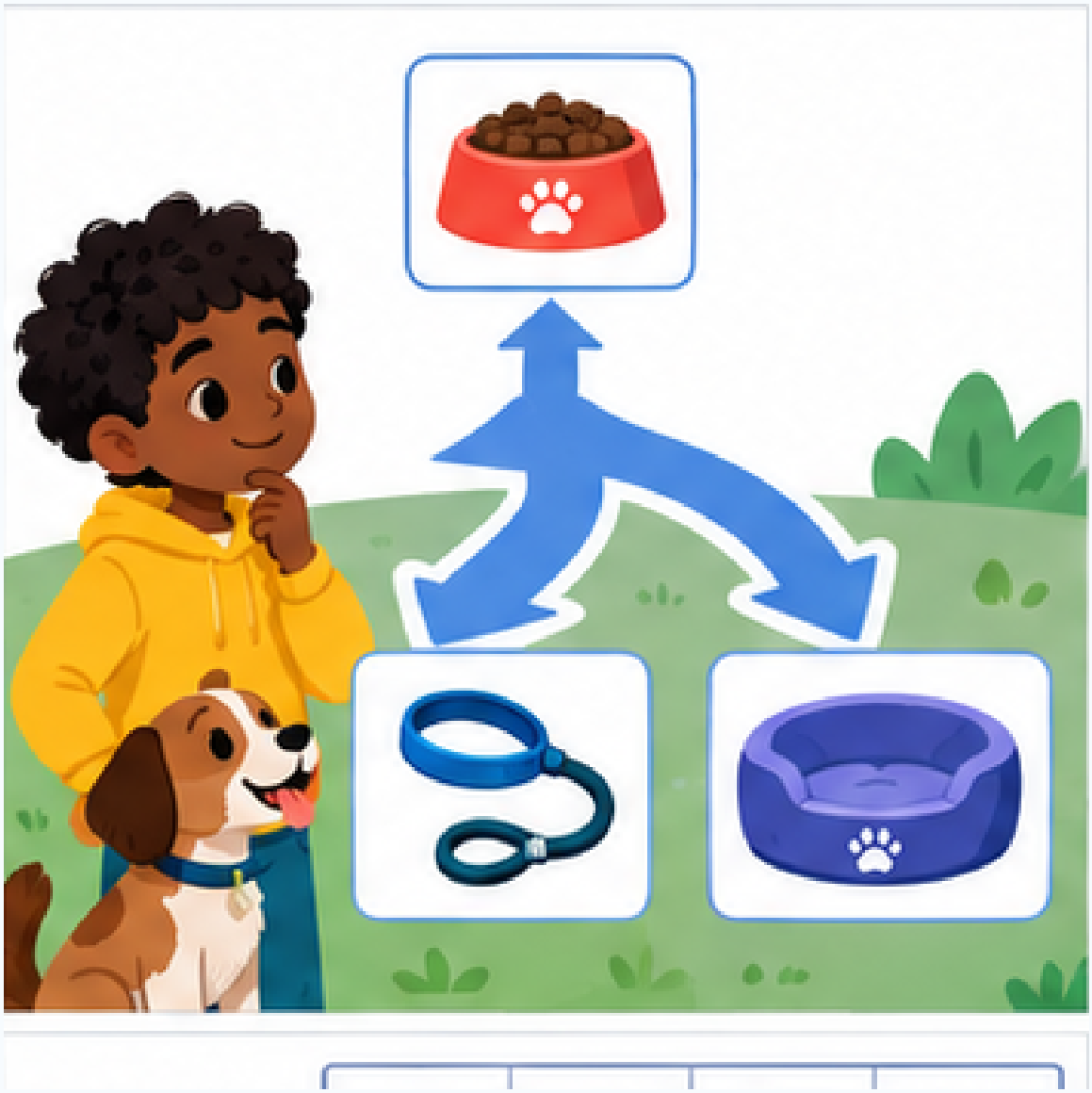
Your learning target

Remember before reading

Without opening the old lesson, explain one key idea from Decompose a Goal. Give an example and a mistake that the idea helps you avoid.

Week 3: **Branch**ing Logic

Ordered **condition**s determine which path a system takes.



THINKER MISSION

Build and trace a three-**branch** pet-care algorithm.

Thinker name and date:

> **Write or draw your reasoning and evidence.**

Readiness for the connected explanation

This week’s end goal

By the end, I can **order branch**es so each **input** receives exactly its intended category.

MAIN LESSON · SECTION 2

Specify the integer-duration sorter before coding

> The main example calls zero Short and uses a twelve-minute boundary.

The complete main classification rule

A fictional activity sorter labels a supplied integer duration. Negative values are Invalid; values from zero through 12 are Short; larger values are Long. The labels describe duration only.

Write this requirement before the program so that a test has an expected answer independent of the code.

Read if, elif, else, colons and indentation

An if statement selects a block when its **condition** is true. An elif is considered only if earlier **condition**s in the same chain were false. An else handles the remaining cases. Colons end the **condition** lines.

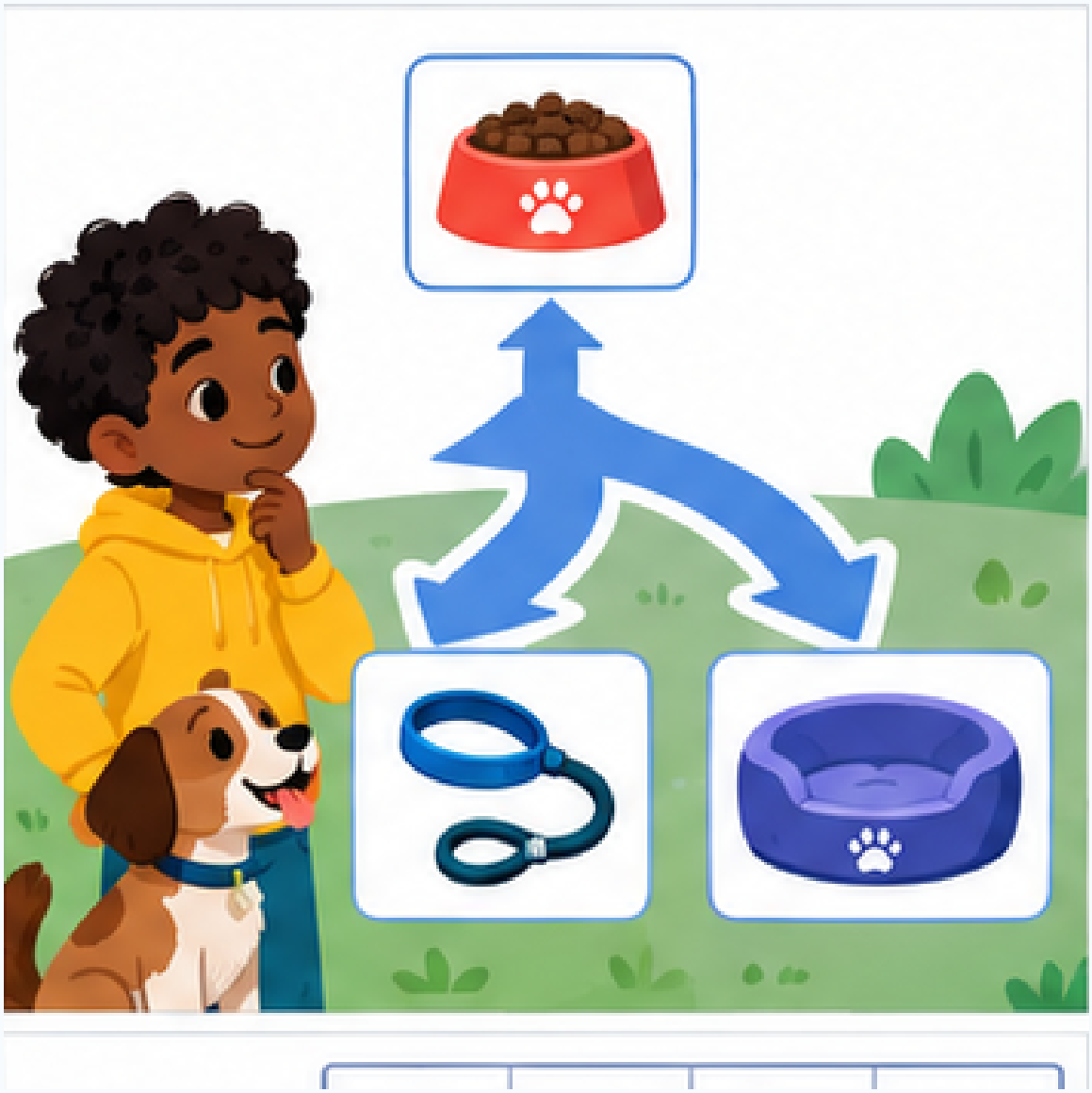
Use four spaces for the indented statements and keep related **branch**es aligned.

▼ **Earlier wording, alternatives and picture transcripts**

PICTURE STORY

Notice → Predict → Explain

Begin with visible **evidence**. Do not explain more than the picture and information support.



MYSTERY

The learners must use **branch**ing logic to solve a problem. What information do they have? What decision or calculation do they make? What would count as **evidence** that it worked?

Three **observation**s (what you can point to):

> **Write or draw your reasoning and evidence.**

My **prediction** and the clue supporting it:

> **Write or draw your reasoning and evidence.**

A closer explanation

MAIN LESSON · SECTION 3

Trace the complete main Python chain

> Each path assigns label before the single unindented print.

Inspect three representative paths

One chain chooses one label

INPUT	CHECKS IN ORDER	SELECTED LABEL
-1	Below zero is True	Invalid
12	Below zero False; <= 12 True	Short
13	Both checks False	Long

Later branches in this chain are skipped after a condition succeeds.

Run the complete main script

```
minutes = 12
if minutes < 0:
    label = "Invalid"
elif minutes <= 12:
    label = "Short"
else:
    label = "Long"
print(label)
```

Explain why exactly one branch is selected

- The **output** is Short. For 12, the first **condition** is false and the second is true. The else block is skipped. print(label) is not indented, so it runs after the selected **branch**.

  Each path assigns label before this line reads it. Change only minutes and save before each fresh run.

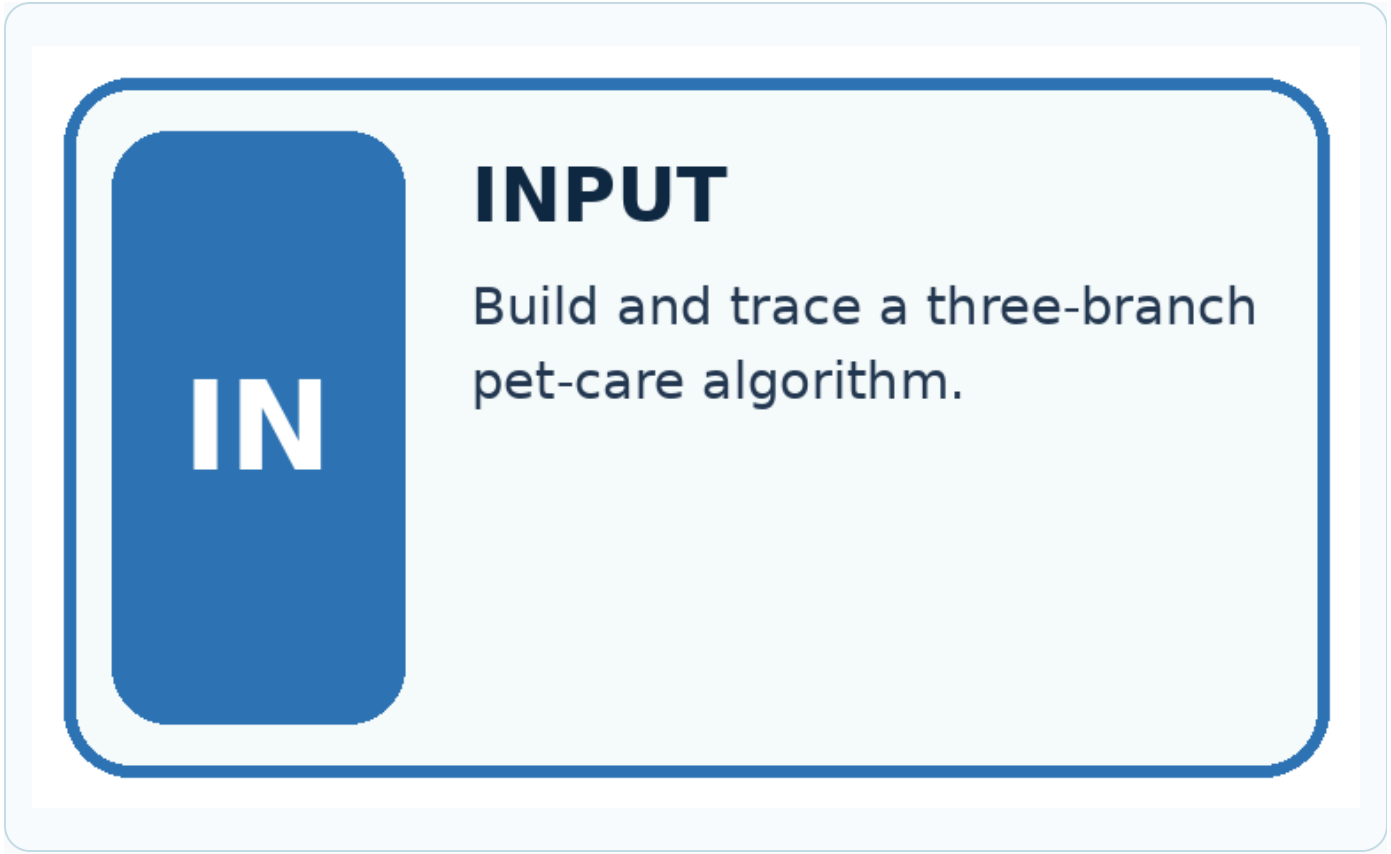
- Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ Earlier wording, alternatives and picture transcripts

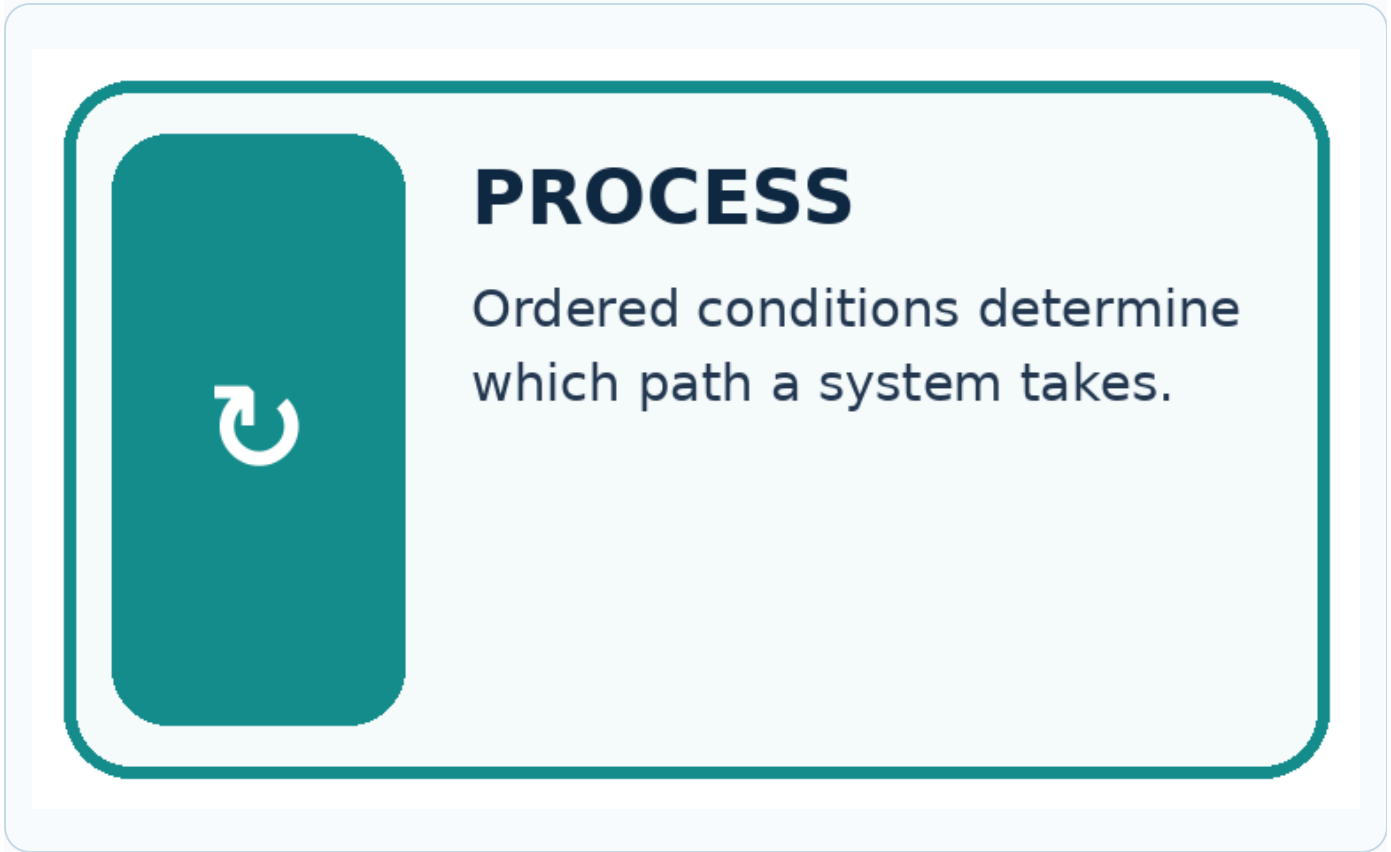
VISUAL **MECHANISM**

Input → **Process** → **Output**

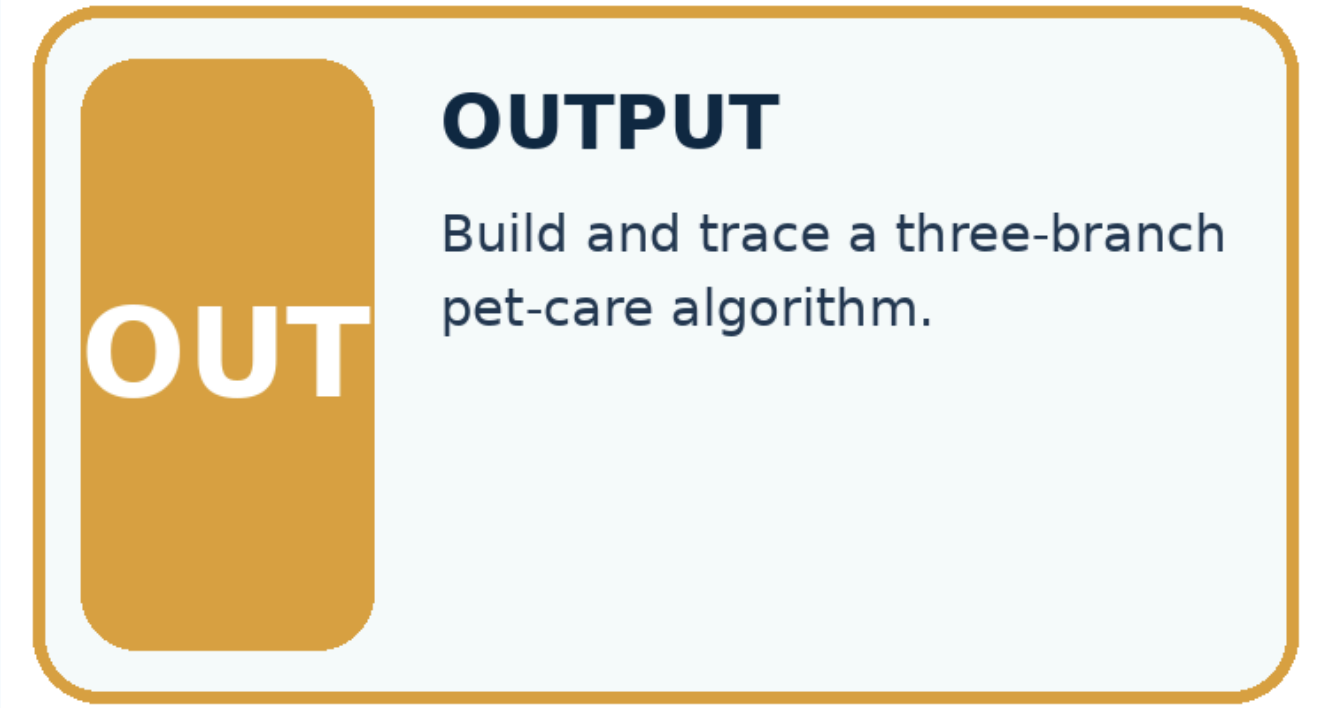
Trace the information. At each stage, ask what changed and how you could test it.


IN
INPUT
Build and trace a three-branch pet-care algorithm.

↓


PROCESS
Ordered conditions determine which path a system takes.

↓



Which stage could fail even when the other two are correct? Explain:

> **Write or draw your reasoning and evidence.**

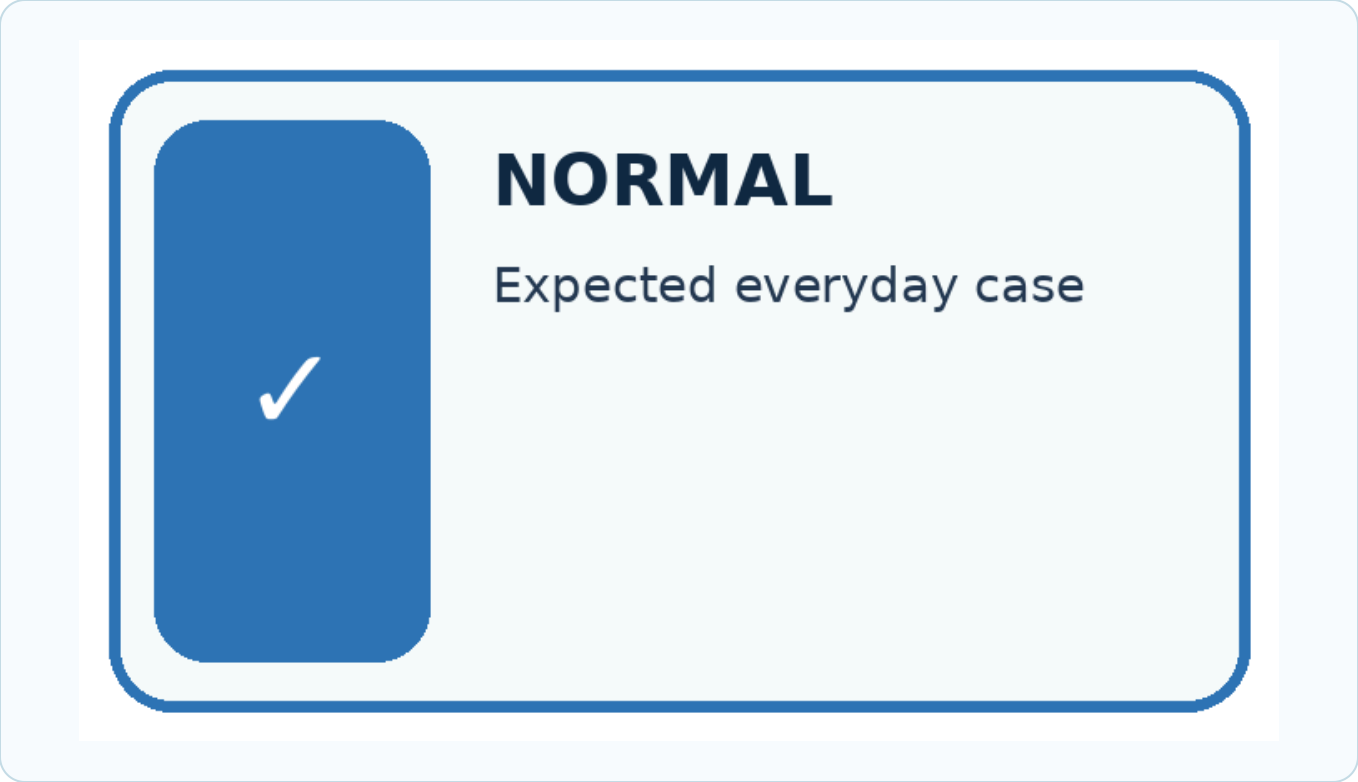
A closer explanation

Connect the picture and the explanation

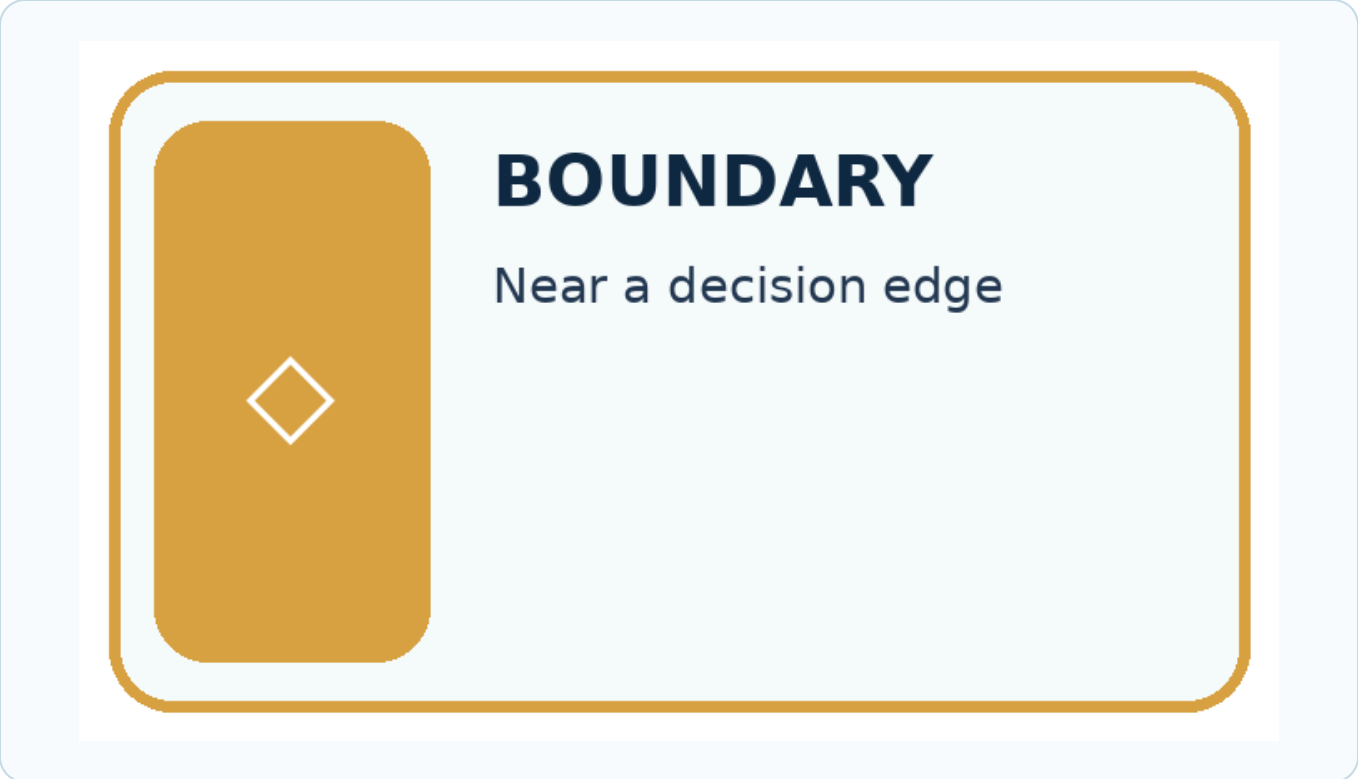
Compare routine, equality, missing and transfer cases

> Branch coverage alone does not establish correct category boundaries.

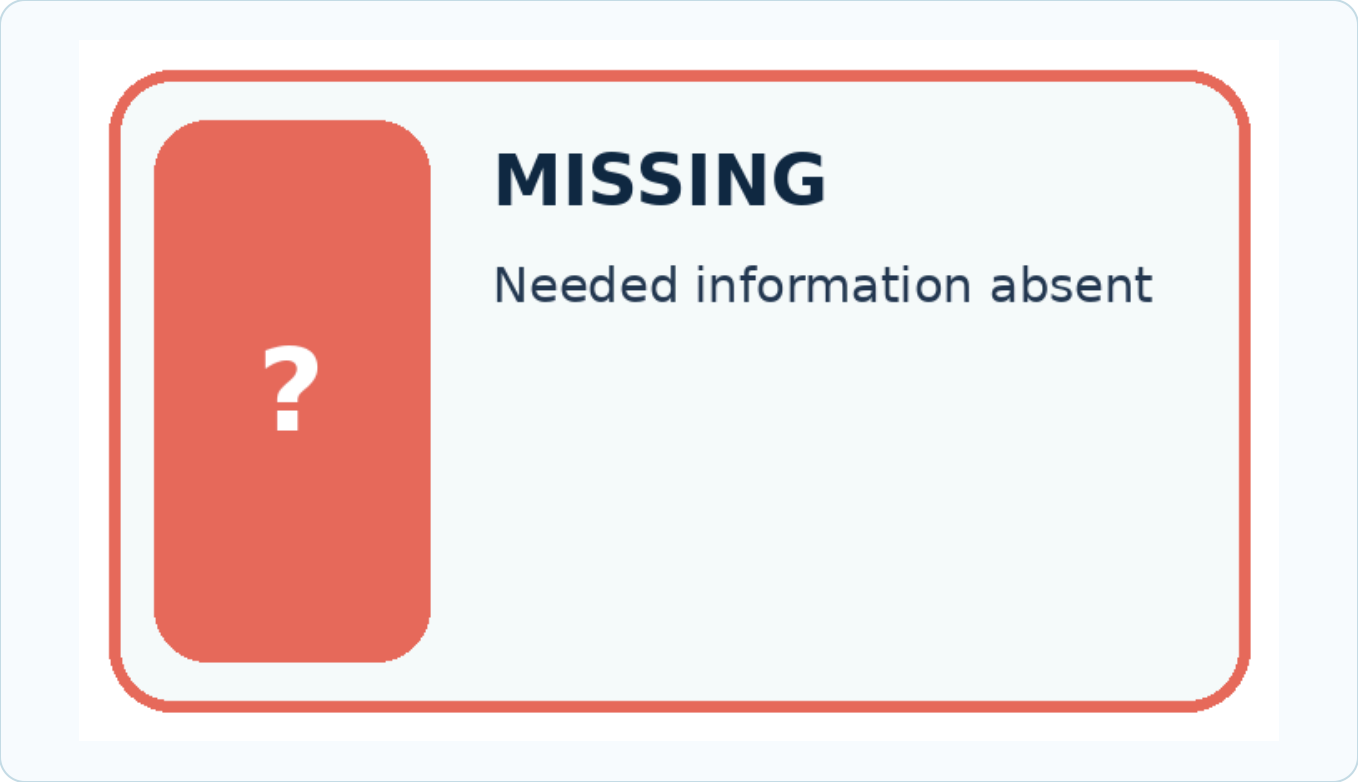
Four pictures with a purpose



Normal case: predict the result and deciding evidence.



Boundary case: identify the exact limit.



Missing information: name what must be clarified.



Transfer case: explain which procedure still applies.

Choose a revealing case

- Which case is most likely to reveal a hidden assumption? Why?

> **Write or draw your reasoning and evidence.**

Predict the supplied changed case

- Change: Move minutes<=10 to the first **branch**.

▼ Check after predicting

Prediction: –1 and 0 wrongly receive Short.

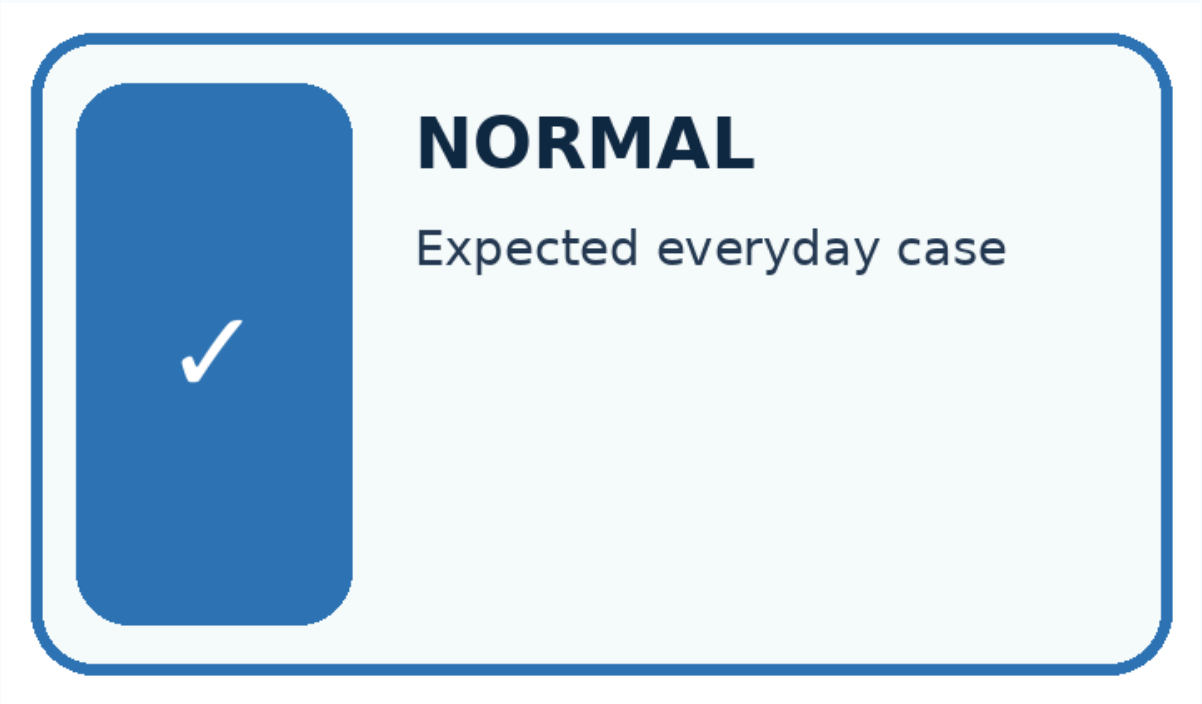
Reason: Both satisfy the earlier comparison before their intended checks are reached.

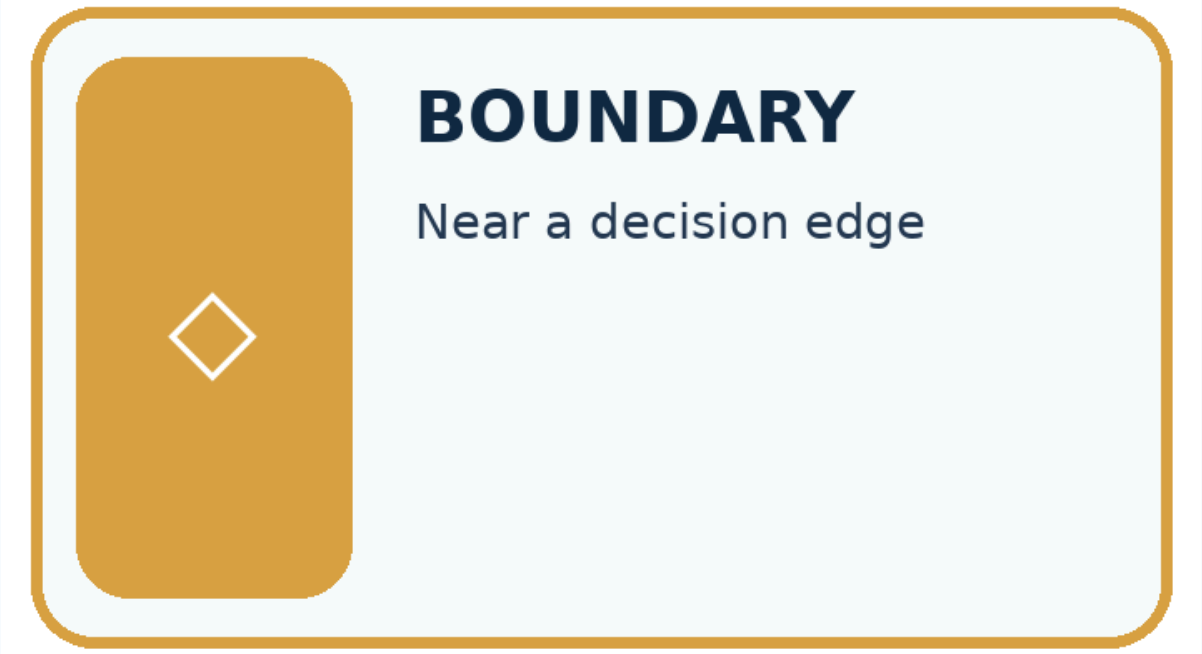
▼ Earlier wording, alternatives and picture transcripts

CASE GALLERY

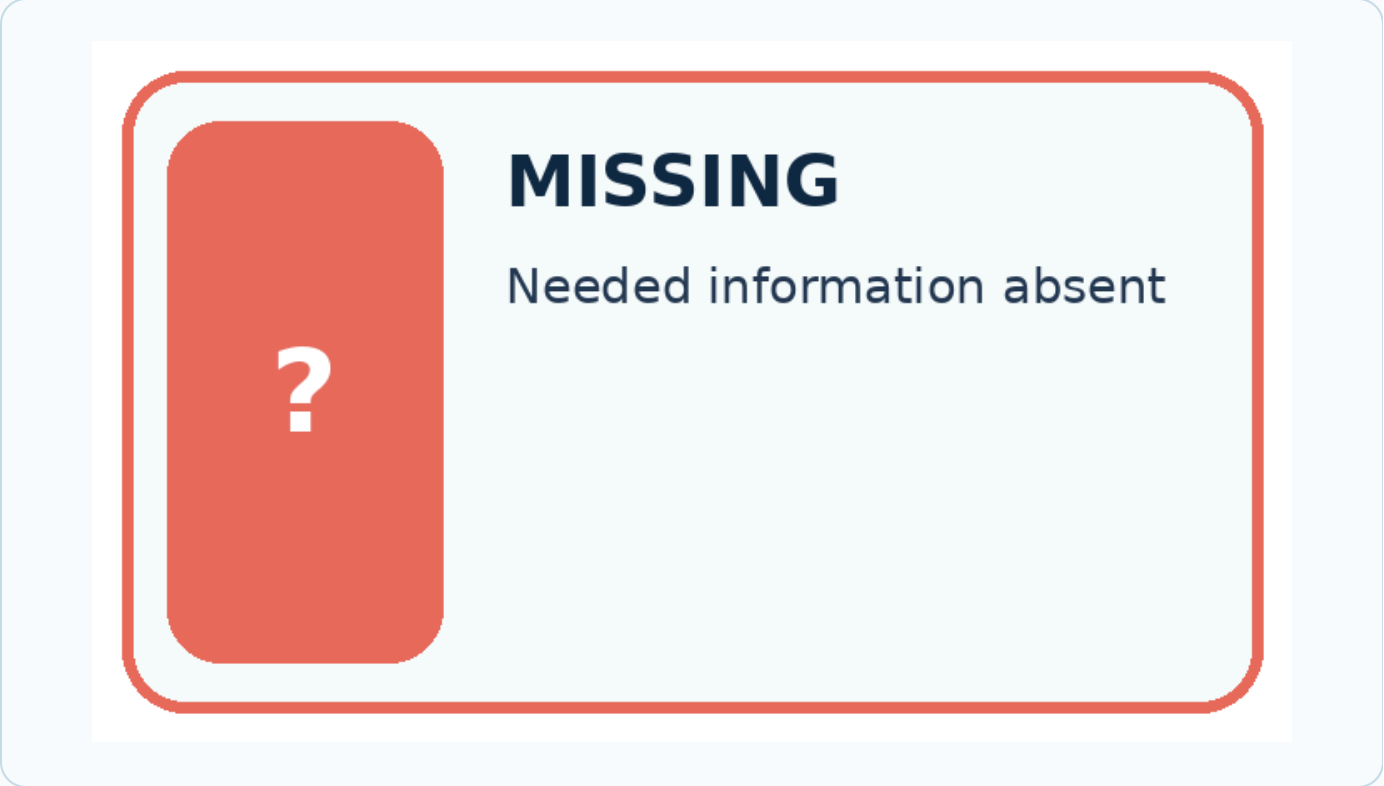
Four Tests, Four Kinds of **Evidence**

A system is not understood until it survives more than one comfortable example.





NORMAL CASE: predict and name **evidence**. **BOUNDARY** CASE: predict and name **evidence**.





MISSING INFORMATION: predict and name **evidence**. TRANSFER CASE: predict and name **evidence**.

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Test the main boundary, then the separate reading-time chain

> The second example has Empty at zero and a ten-minute boundary; do not reuse the main rule’s labels.

Plan main tests below, at and above twelve

Input minutes	Expected label	Reason for this test
-1	Invalid	Negative input rejection
0	Short	Lowest valid duration
11	Short	Just below the upper boundary
12	Short	Exact upper boundary
13	Long	Just above the boundary

These tests exercise every **branch** and the equality case. A change from <= 12 to < 12 is exposed by 12; testing 11 alone misses it.

Branch coverage means every **branch** was exercised, but it does not prove every **condition** is correct.

Predict the strict-comparison defect

> **Suppose the program uses < 12 instead of <= 12. What is printed for 12, why does that violate the written rule, and which earlier test should still pass after repair?**

Inspect the main twelve-minute diagram

AI ACADEMY / AI-2/3 · 03 · INSPECT THE LESSON MODEL

One chain chooses one label

Input	Checks in order	Selected label
-1	Below zero is True	Invalid
12	Below zero False; <= 12 True	Short
13	Both checks False	Long

Later branches in this chain are skipped after a condition succeeds.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the separate reading-time categories

A fictional reading-time labeler uses one Python if/el**if/else** chain.

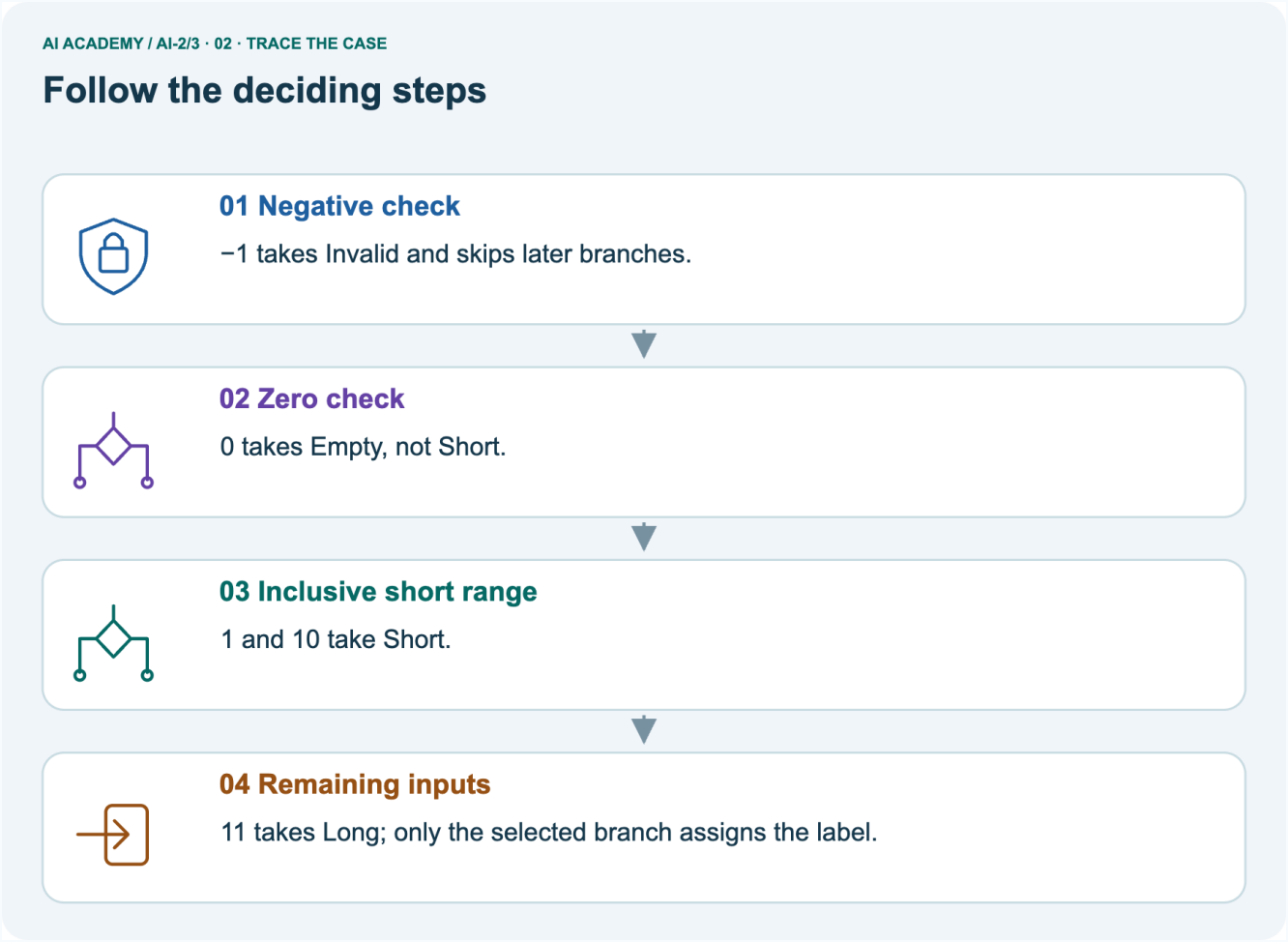
Whole minutes below 0: Invalid; 0: Empty; 1–10: Short; above 10: Long.

Order checks as minutes<0, minutes==0, minutes<=10, otherwise. Print the assigned label once.

```
minutes = 0
if minutes < 0:
    label = "Invalid"
elif minutes == 0:
    label = "Empty"
elif minutes <= 10:
    label = "Short"
else:
```

```
    label = "Long"
print(label)
```

Trace order, equality and skipped branches



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Negative check	–1 takes Invalid and skips later branches.
2. Zero check	0 takes Empty, not Short.
3. Inclusive short range	1 and 10 take Short.
4. Remaining inputs	11 takes Long; only the selected branch assigns the label.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

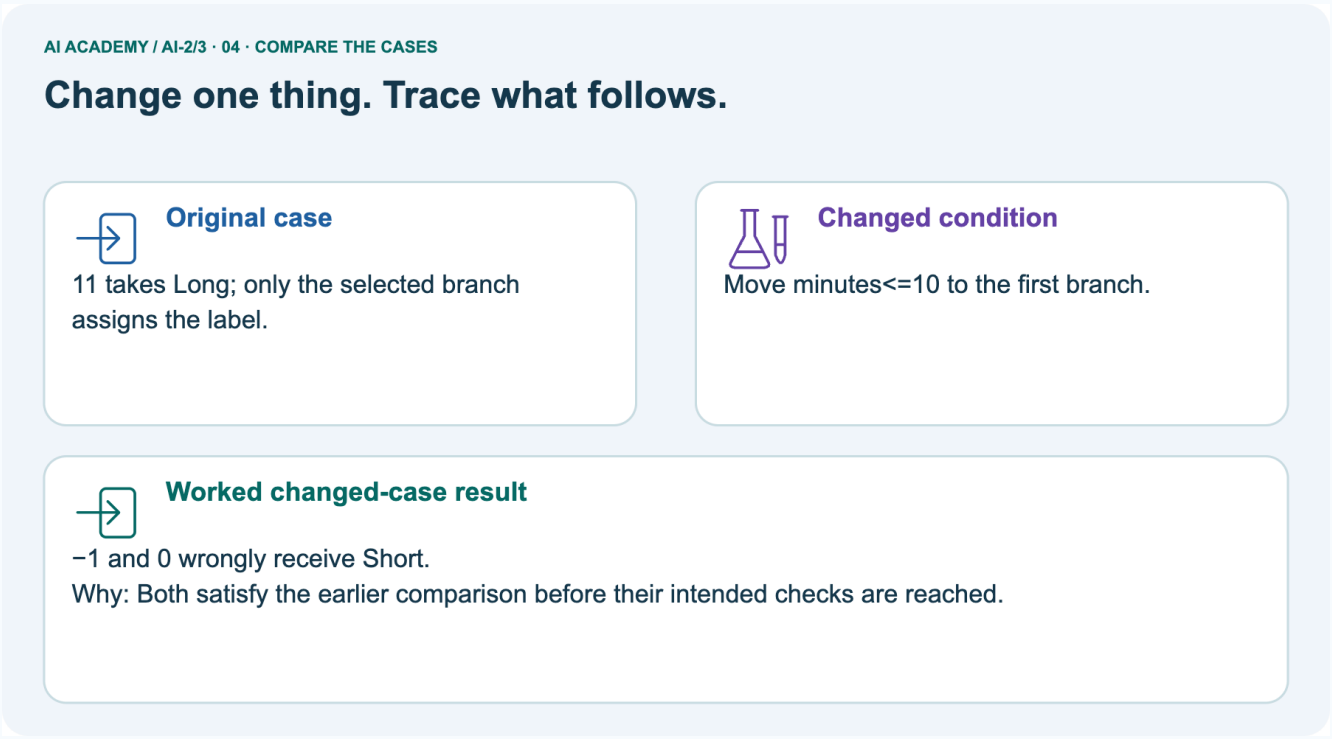
The third comparison alone would include negative values and zero, but earlier **branch**es already handle them. An elif chain selects its first true **branch**.

Multiple independent if statements would have different behavior and could overwrite an earlier label. **Boundary** traces show why **order** belongs to the specification.

Predict the reordered chain

- Before opening the reordered-branch picture, move the reading-time check minutes<=10 before the negative and zero checks in the same chain. Predict the labels for -1 and 0, and identify the intended categories that become unreachable for those inputs.

▼ Check the branch-order failure



Follow the labelled picture with the matching explanation. Open the vector diagram

State the new rule's expectations and type limit

- Read the labelled picture: Give expected labels for –1,0,10,11 and identify the equality **boundary** for Short.
- The example assumes whole numeric minutes; strings, missing values and fractional values need a separate declared **input** contract.

▼ Earlier wording, alternatives and picture transcripts

WORKED **EVIDENCE**

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.

WEEK 3 • WORKED EVIDENCE

1. IF pet is hungry → feed.
2. ELSE IF pet needs exercise → walk or play.
3. ELSE → provide a quiet rest space.
4. Boundary test: what if hungry and needs exercise? The first true branch wins.

1. IF pet is hungry → feed.

2. ELSE IF pet needs exercise → walk or play.

3. ELSE → provide a quiet rest space.

4. **Boundary** test: what if hungry and needs exercise? The first true **branch** wins.

Recalculate, retrace, or restate the reasoning in your own representation:

> **Write or draw your reasoning and evidence.**

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

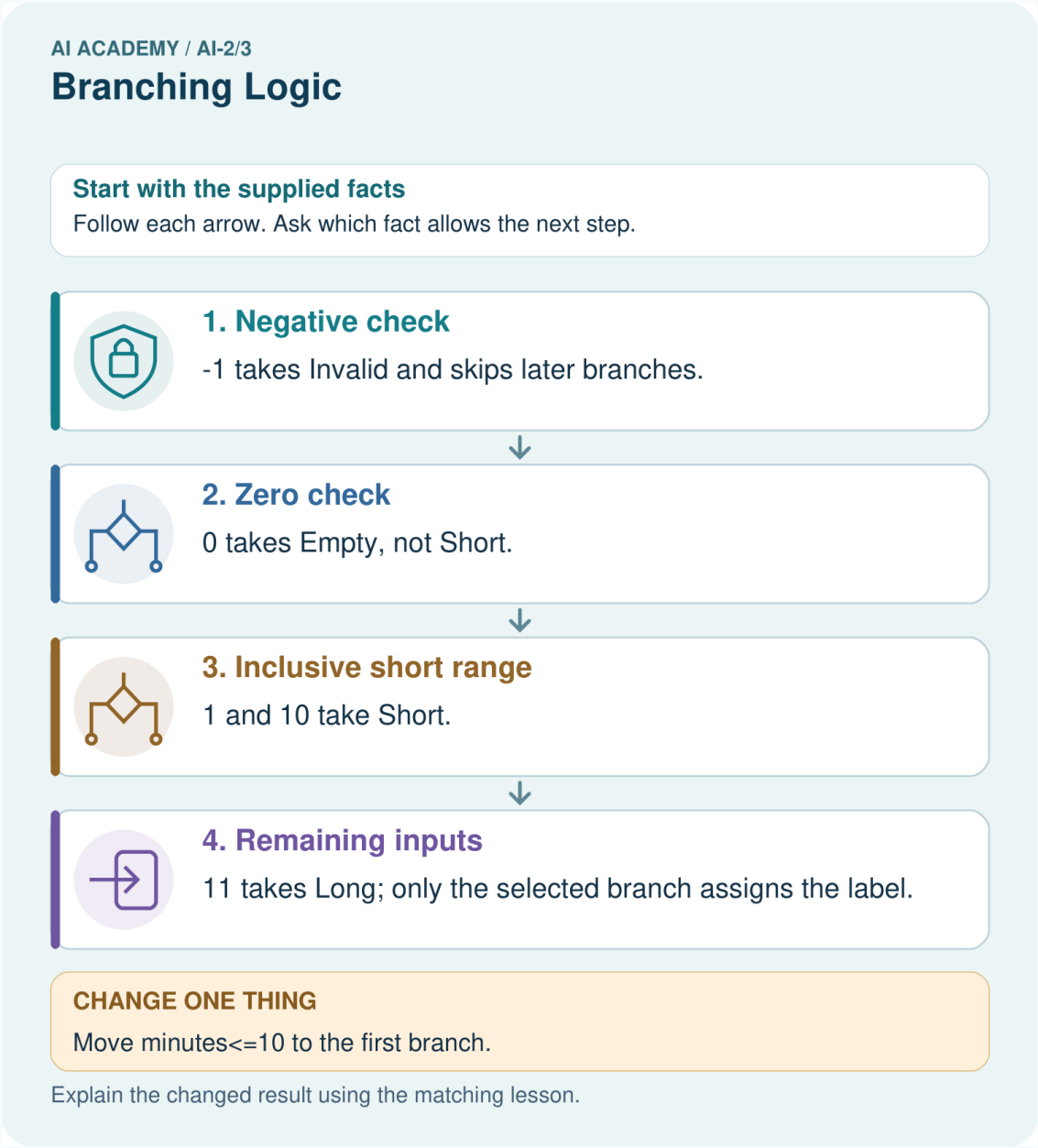
A closer explanation

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Branching Logic

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing: Move minutes<=10 to the first **branch**. Predict the result before checking the explanation. What remains the same?

MAIN LESSON · SECTION 7

Record checked and skipped conditions honestly

> Coverage needs both paths and revealing boundary values.

Keep a per-run trace and regression tests

For each run, record **input**, expected label, **condition**s actually checked and observed label. Mark skipped **condition**s as skipped rather than pretending the program evaluated them. After a repair, rerun earlier cases as well as the failing one. Weeks 4 and 5 trace repeated blocks and examine loop progress and stopping.

Challenge three-branch coverage as proof

> **A learner tests -1, 2 and 20 and concludes the sorter is completely correct because all three branches ran. Explain what important case is missing and why branch coverage alone is insufficient.**

▼ Earlier wording, alternatives and picture transcripts

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak **evidence**.

!

COMPARE • DIAGNOSE

Weak idea: All true conditions run even in an if/else-if chain. Strong move: locate evidence, change one cause, and retest.

Evidence record	Expected	Actual	Likely cause / next test
normal case	works		
boundary case	stated rule		
missing input	safe response		
fresh transfer	generalizes		

Repair one cause. Why is this repair better than changing several things?

Write or draw your reasoning and evidence.

A closer explanation

Practise: find the assignment that overwrites the label

> Separate if statements are not an elif chain.

Run the faulty two-if script

```
minutes = -1
if minutes < 0:
    label = "Invalid"
if minutes <= 12:
    label = "Short"
else:
    label = "Long"
print(label)
```

Locate the overwrite and attached else

This faulty version prints Short. The second if starts a separate decision and overwrites Invalid. Its else belongs to that second if. Restoring elif makes the second test depend on the first test failing.

Moving print inside one **branch** can also hide **output** on other paths; **indentation** affects behavior, not just appearance.

Keep the integer-only scope explicit

The script rejects negative integers, but it does not validate arbitrary types or check a measured duration. A supplied text value can raise an error during comparison.

Keep the stated integer **input** contract and report the limitation honestly.

Trace main paths in W2

Use workbook W2 for the connected example “**Branch**ing and **boundary** tests”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Repair the every-condition-runs claim

- Use the worked case for **Branch**ing and **boundary** tests. Proposed explanation: “Every **condition** in the chain always runs”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Return to the reading-time boundary

Give expected labels for –1,0,10,11 and identify the equality **boundary** for Short.

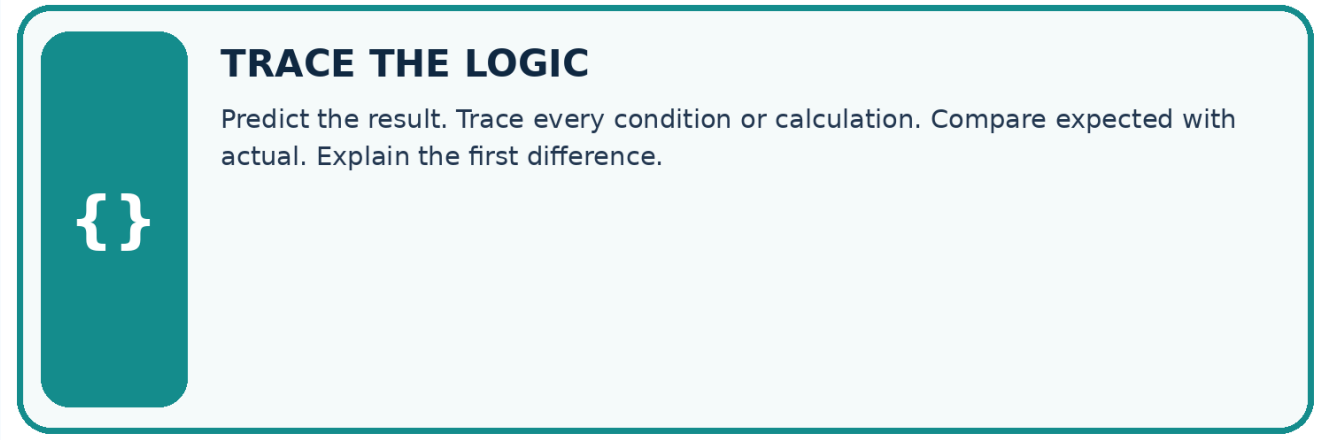
This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ Earlier wording, alternatives and picture transcripts

LOGIC STUDIO

Trace Before You Run

Block code, pseudocode, tables, and diagrams are different ways to expose the same reasoning.



SET **evidence**_record PREDICT expected_result RUN one normal case and one **boundary** case COMPARE actual_result with expected_result IF different: locate first cause; change one factor; RETEST EXPLAIN result, limitation, and next safe step

Trace one **input**. Which line changes the state or chooses a path?

Write or draw your reasoning and evidence.

A closer explanation

Practise with less help: Trace the tested path

Check your reasoning

Explain the picture

My explanation using a pictured step and the given facts

MAIN LESSON · SECTION 9

Investigate order and exact boundaries with supplied cases

> Keep the paper token rule distinct from the integer-duration code.

Prepare and compare the six fictional token cards

Paper branch station: use fictional pet cards, not real animal-care decisions. The supplied rule is Hungry Yes→Feed token; otherwise Needs play Yes→Play token; otherwise Rest token. Cards A=(Yes,Yes), B=(Yes,No), C=(No,Yes), D=(No,No), E=(Yes,Yes), F=(No,Yes). Predict each token under this ordered chain, then swap the first two checks and predict which cards change. Explain the first true condition; these tokens are an algorithm exercise, not care instructions.

Predict, run and repair the supplied scripts

1. Code station: predict and run the main duration script separately for -1,0,11,12,13. Record expected label, conditions actually tested, skipped conditions and actual output/error. Compare the two-independent-if version at -1, repair its second if to elif, then rerun earlier cases. Change <=12 to <12 in a separately labelled version and test the exact boundary. Keep the independent tile sorter for an uncoached attempt and label paper-only results as traces.

Keep the evidence table

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

> **Test / input:** normal
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____

> **Change or keep?:** _____

> **Test / input:** boundary
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

> **Test / input:** missing
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

> **Test / input:** fresh transfer
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

Explain the most revealing test

- What did the hardest test teach you that the normal test did not?

- Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

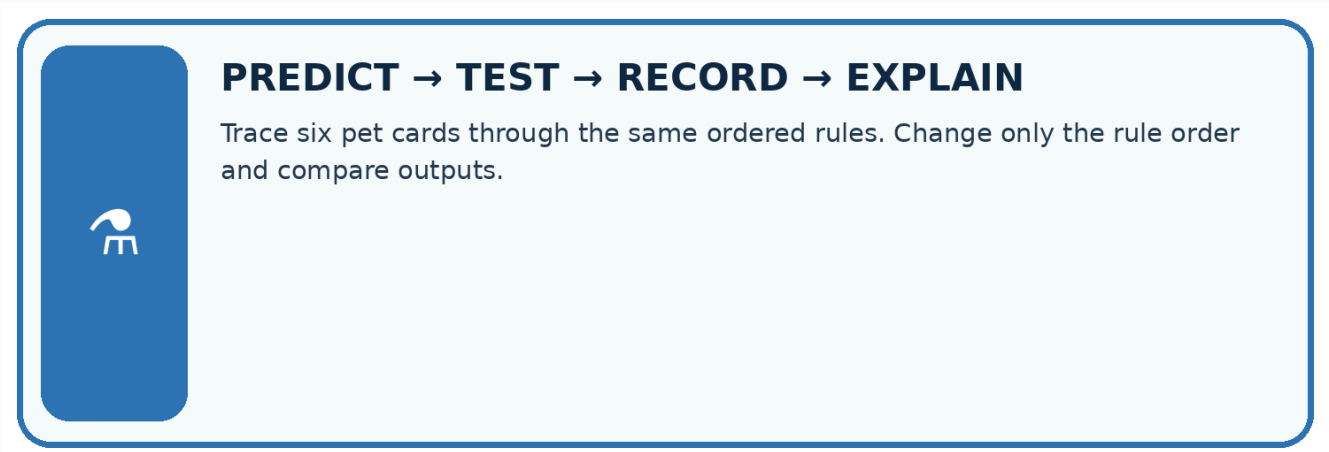
Write or draw your reasoning and evidence.

▼ **Earlier wording, alternatives and picture transcripts**

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



Trace six pet cards through the same **order**ed rules. Change only the rule **order** and compare **output**s.

Plan → try → check

Try alone: implement the different tile categories

> Use its zero category and six-tile boundary before feedback.

The independent category contract

Use an integer tiles count. Label a negative value Invalid, zero Empty, 1 through 6 Small and values above 6 Large. Write an **order**ed if elif else script and predict -1, 0, 1, 6 and 7. Include one print after the chain.

Keep your first attempt

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Use feedback afterwards

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ Earlier wording, alternatives and picture transcripts

Independent case: Build a new sorter independently

Week 3 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

If pages are at most 80, choose short; otherwise choose long. Classify books with 79, 80 and 81 pages and explain the **boundary**.

My answer, with a trace, calculation or supporting **evidence**

> **Write or draw your reasoning and evidence.**

Create and test

Change one relevant detail. Predict the result before testing. Include a **boundary** or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

Connect to your project

Branch when a reader has a hard constraint such as language, format, or maximum length.

The **evidence** I will keep

> **Write or draw your reasoning and evidence.**

Transfer practice from the original programme

Classify lengths with if pages<80 then Short; else if pages<=160 then Medium; else Long. Trace 79,80,160,161. What happens if the <=160 **branch** is placed first?

> **Explain the mechanism, evidence and one limit.**

Use feedback to improve

MAIN LESSON · SECTION 11

Add explicit eligibility and clarification branches

> Keep hard constraints separate from later scoring and human choice.

This week’s bounded contribution

For the Transparent Book Recommender, specify hard constraints and invalid or missing inputs before ranking. Use fictional records and the declared project rules. A preference score must not silently override an eligibility condition. This week contributes explicit branches and reasons; it does not establish reader satisfaction or a complete recommender.

The checkpoint and system map

THIS WEEK'S CHECKPOINT **Branch** when a reader has a hard constraint such as language, format, or maximum length.

SYSTEM MAP Reader need → allowed **input**s → transparent rule or score → ranked candidates → explanation → test record → **human decision**

Validate recommender requests before ranking items, retaining explicit labels for missing or invalid cases.

Record the branch and who may act

- What will you add, change, or verify in the project this week?
- What should remain under **human control**?

Write or draw your reasoning and evidence.

Write or draw your reasoning and evidence.

Preserve privacy and uncertainty

RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits. Never confuse a ranking with a fact about a person.

▼ **Earlier wording, alternatives and picture transcripts**

CUMULATIVE PROJECT

Transparent Book Recommender

Every week adds one inspectable piece. The system must show how it reached a result.



Evidence for your project

MAIN LESSON · SECTION 16

Finish: explain the selected path and a boundary test

> Close your notes and say why later conditions were skipped.

Your exit explanation

- Close your notes. Explain this goal in your own words: Trace the selected **branch** in an if elif else chain

My deciding step, evidence and one remaining question

Choose the next revealing test

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Your quick word guide**

REFERENCE · SECTION 5

Your quick word guide

> Use the words to explain a particular step, record or limit.

Core vocabulary

Word	My own example	Not this / boundary
condition	A statement evaluated as true or false to help select an action.	
branch	One path taken when a condition selects among alternatives.	
if/else	A programming structure that chooses a path according to whether a condition is true or false.	
order	The sequence in which steps are performed or items are read.	

Terms for the connected explanation

Word	Meaning
Indentation	Leading spaces that identify a Python code block
Boundary	A value where a classification may change

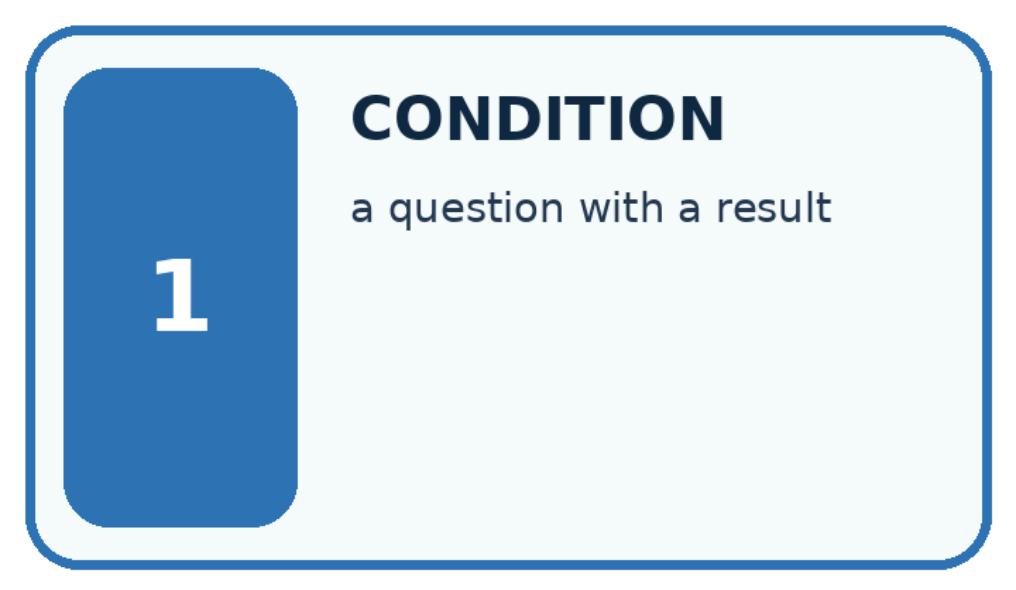
Word	Meaning
Regression check	A previous test rerun after an edit to detect a new failure

▼ Earlier wording, alternatives and picture transcripts

VOCABULARY LAB

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.



2

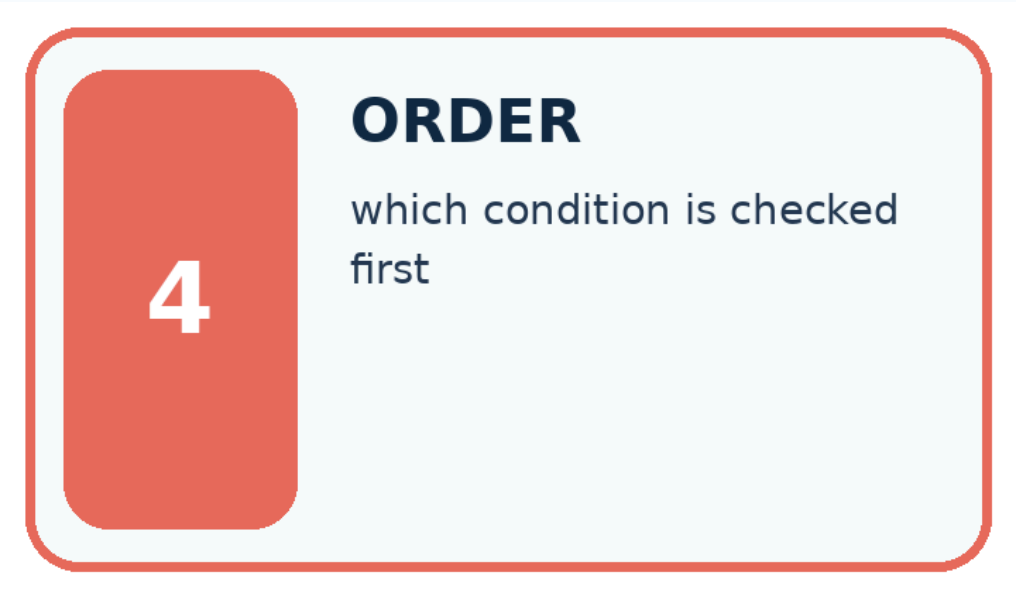
BRANCH

one path chosen after a condition

3

IF/ELSE

a two-way decision structure



Words used in the closer explanation

▼ **Choose a stronger thinking challenge**

REFERENCE · SECTION 10

Choose a stronger thinking challenge

> Choose the next useful challenge rather than treating the whole ladder as compulsory.

Pick your next step

- 1 • NAME: Define the concept without repeating the textbook.
- 2 • TRACE: Show the information, state, **branch**, calculation, or **evidence** path.
- 3 • APPLY: Solve a different example independently.
- 4 • CHALLENGE: Find a **boundary** case, counterexample, or unfair outcome.

- 5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

- My strongest **evidence** of independent understanding:

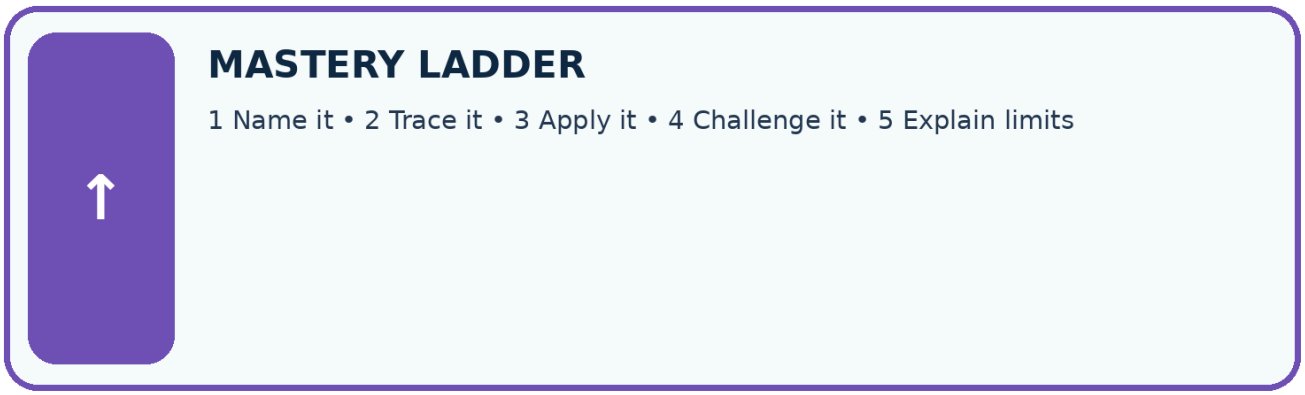
Write or draw your reasoning and evidence.

▼ Earlier wording, alternatives and picture transcripts

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the worked example.



▼ Reference: check your explanation

Reference: check your explanation

> Use after practice. Build a new sorter independently is the selected independent case.

Check the evidence and its limit

- 1. My fresh example, trace, calculation, or design:
- 2. My **evidence** that it works (include a **boundary** or missing-data case):
- 3. One limitation, fairness concern, or situation needing human review:

Write or draw your reasoning and evidence.

Write or draw your reasoning and evidence.

Self-check and record

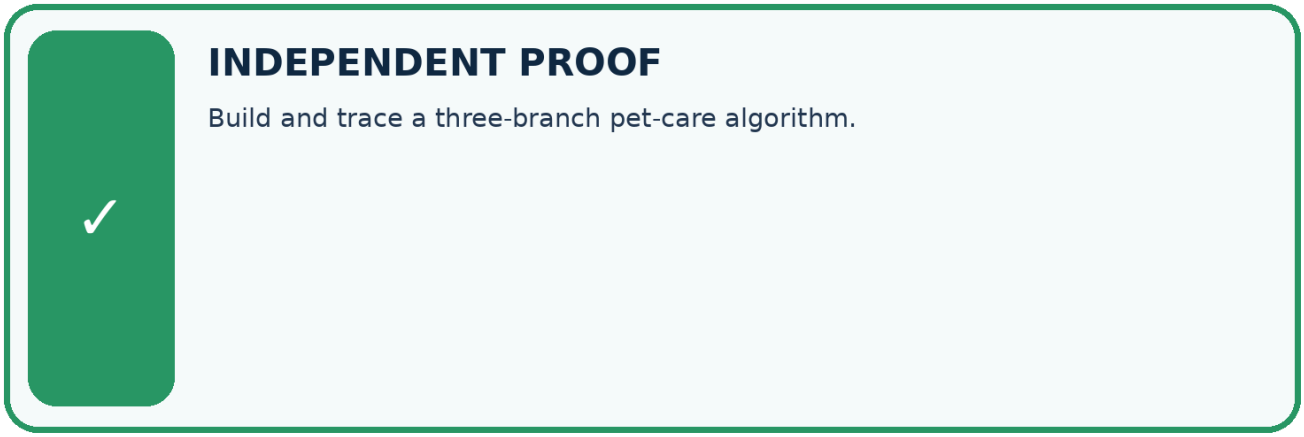
SELF-CHECK ☐ I explained the **mechanism** ☐ I used **evidence** ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

▼ **Earlier wording, alternatives and picture transcripts**

INDEPENDENT MASTERY PROOF

Checklist for explaining this week's idea

Use after practice. Build a new sorter independently is the selected independent case.



EXPLANATION CHECK Trace only the conditions reached, preserve ordered **branch**es and test equality against the written category rule.

> **Write or draw your reasoning and evidence.**

▼ **Extra practice: Test strict and inclusive page boundaries**

REFERENCE · SECTION 13

Extra practice: Test strict and inclusive page boundaries

> Try the question before opening its worked explanation.

Read the supplied case and question

If pages<100 choose Short; otherwise if pages<=200 choose Medium; otherwise Long.

Classify 99,100,200,201 pages.

Your explanation

- Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

▼ Check after trying

Short, Medium, Medium, Long. Strict and inclusive boundaries and **branch order** matter.

Evidence or stage	Value or result
99 pages	Less than 100 → Short
100 pages	100 to 200 inclusive → Medium
200 pages	100 to 200 inclusive → Medium
201 pages	Greater than 200 → Long

▼ Earlier wording, alternatives and picture transcripts

Week 3 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

The question

Worked explanation

Explain it in your own words

Trace the supplied case visually

Test strict and inclusive page boundaries

If pages<100 choose Short; otherwise if pages<=200 choose Medium; otherwise Long.

Classify 99,100,200,201 pages.

Short, Medium, Medium, Long. Strict and inclusive boundaries and **branch order** matter.

▼ Reference: original lesson checklist

REFERENCE · SECTION 15

Reference: original lesson checklist

> Use an additional checklist when it helps inspect your explanation.

Targets and evidence

By the end, I can	How I will show it
Trace the selected branch in an if elif else chain	Explain the deciding step, show a trace or test, and state one limit.
Use indentation to identify which statements belong to a branch	Explain the deciding step, show a trace or test, and state one limit.
Test values at and around a decision boundary	Explain the deciding step, show a trace or test, and state one limit.

Keep your route and first prediction

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Earlier wording, alternatives and picture transcripts

Start with this question: A fictional reading-time labeler uses one Python if/el**if/else** chain.

▼ Extra practice: pictures, repair and feedback

REFERENCE · SECTION 17

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- A fictional reading-time labeler uses one Python if/el**if/else** chain.
- Which labelled step matches this detail? 11 takes Long; only the selected **branch** assigns the label.
- 1. Negative check
- 2. Zero check
- 3. Inclusive short range
- 4. Remaining **input**s
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Remaining **input**s 11 takes Long; only the selected **branch** assigns the label.

The third comparison alone would include negative values and zero, but earlier **branch**es already handle them. An elif chain selects its first true **branch**.

Multiple independent if statements would have different behavior and could overwrite an earlier label. **Boundary** traces show why **order** belongs to the specification.

Changed-case question

- Changed-case practice: Move minutes<=10 to the first **branch**. Which conclusion is supported by the supplied facts?
- 1. We can decide the result without examining the changed **input** or the procedure.
- 2. –1 and 0 wrongly receive Short.
- 3. This one example guarantees correct results for every future **input**.
- Hint if needed: Use the changed **condition** and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the changed case**

After your attempt, compare with the supported explanation: –1 and 0 wrongly receive Short. Both satisfy the earlier comparison before their intended checks are reached.

Repair and save your work

Review Decompose a Goal (ai-2/2). Goal: Distinguish a function definition from a function call. Short practice: What happens visibly if the caller stores needed but omits print(needed)?.

Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work
</details>

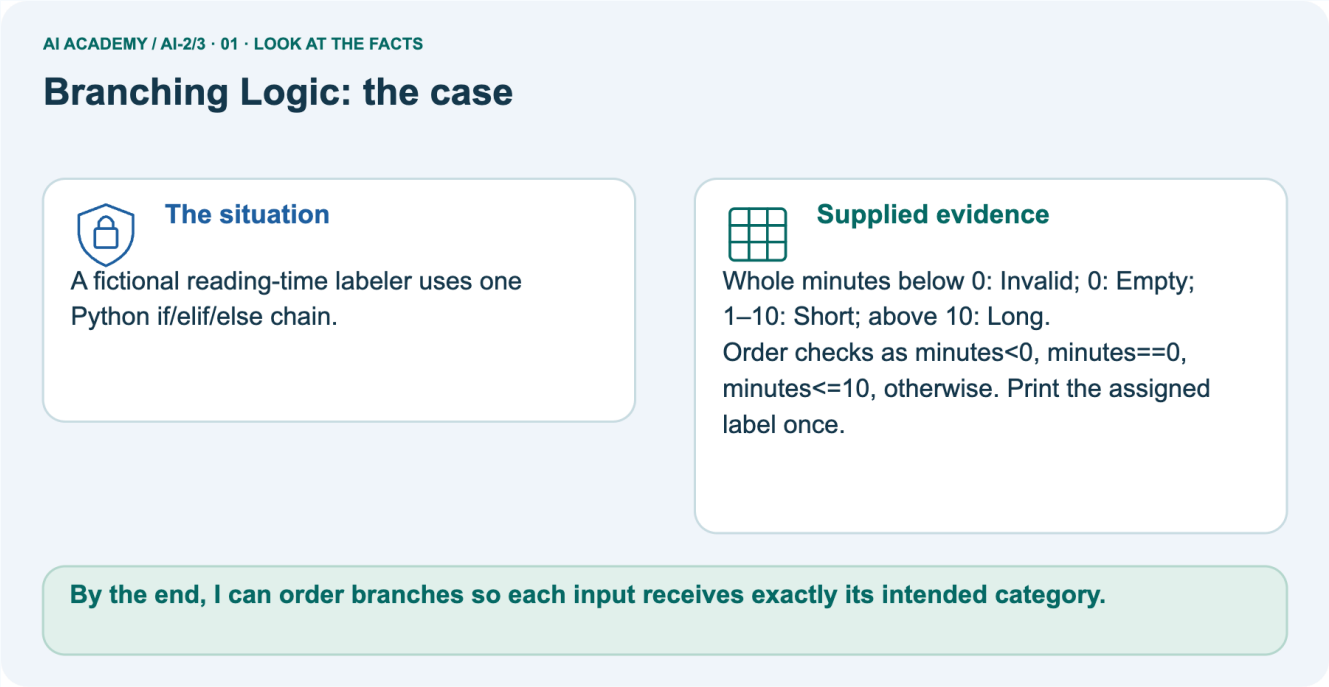
<details>
<summary>Reference: the six-picture journey</summary>
</details>

REFERENCE · SECTION 18

Reference: the six-picture journey

> Revisit the picture that serves the current question; key diagrams already sit beside the main explanation.

01. Branching Logic: the case



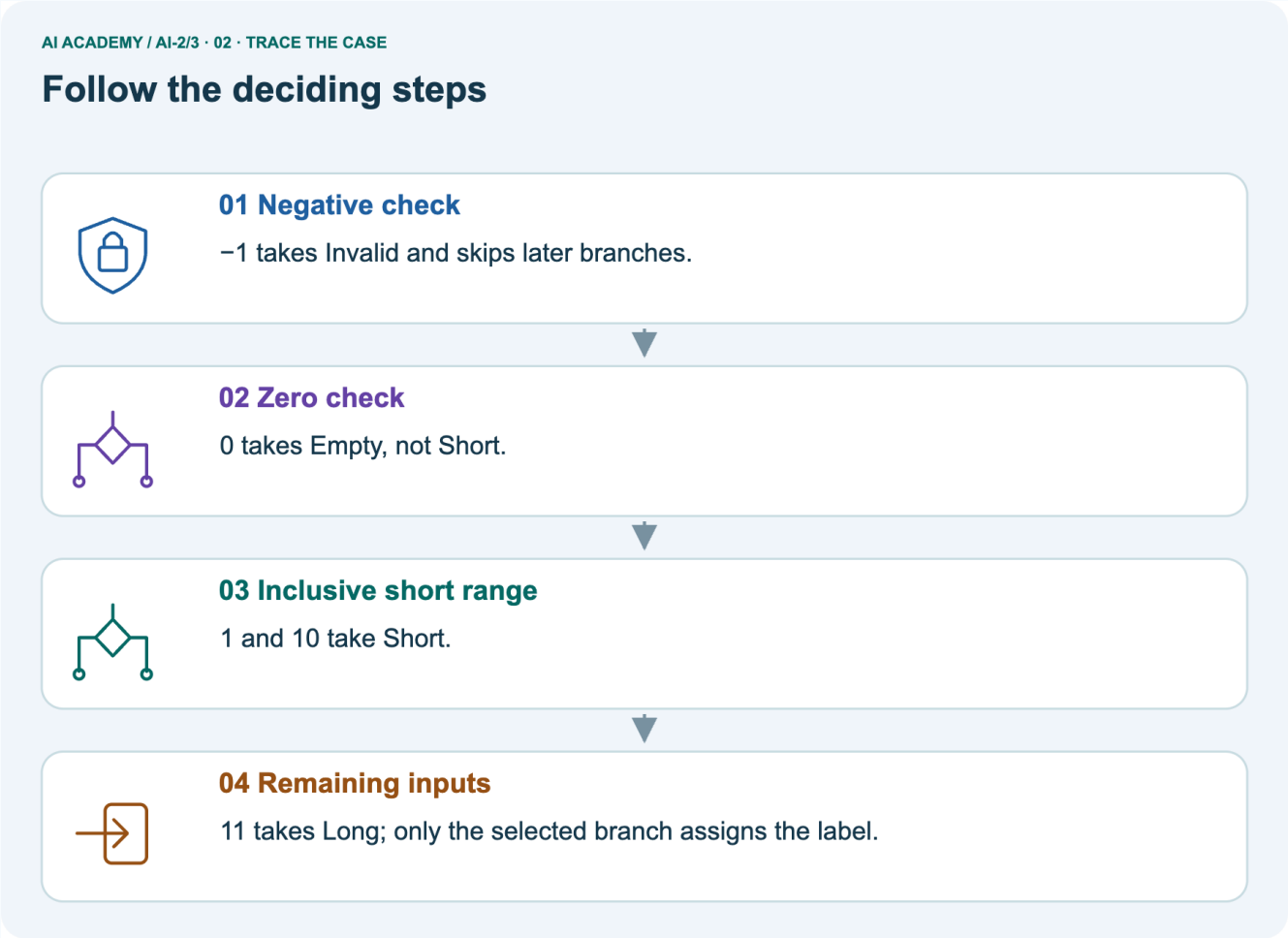
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional reading-time labeler uses one Python if/el**if/else** chain. Supplied facts: Whole minutes below 0: Invalid; 0: Empty; 1–10: Short; above 10: Long. **Order** checks as minutes<0, minutes==0, minutes<=10, otherwise.

Print the assigned label once.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Negative check: −1 takes Invalid and skips later **branch**es. → Zero check: 0 takes Empty, not Short. → Inclusive short range: 1 and 10 take Short. → Remaining **input**s: 11 takes Long; only the selected **branch** assigns the label.

Point to each numbered stage and explain which recorded **input** it uses.

03. One chain chooses one label

AI ACADEMY / AI-2/3 · 03 · INSPECT THE LESSON MODEL

One chain chooses one label

Input	Checks in order	Selected label
-1	Below zero is True	Invalid
12	Below zero False; <= 12 True	Short
13	Both checks False	Long

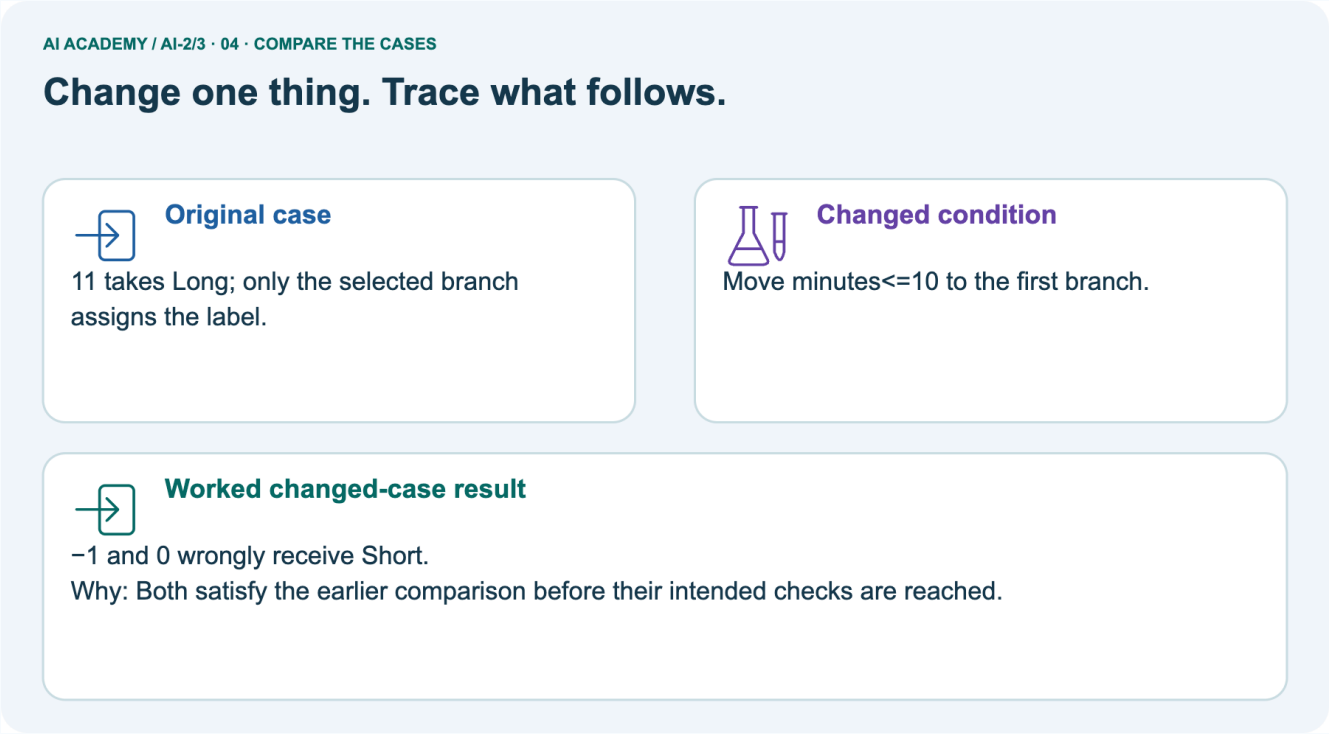
Later branches in this chain are skipped after a condition succeeds.

Follow the labelled picture with the matching explanation. Open the vector diagram

One chain chooses one label. -1 | Below zero is True | Invalid. 12 | Below zero False; <= 12 True | Short. 13 | Both checks False | Long

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



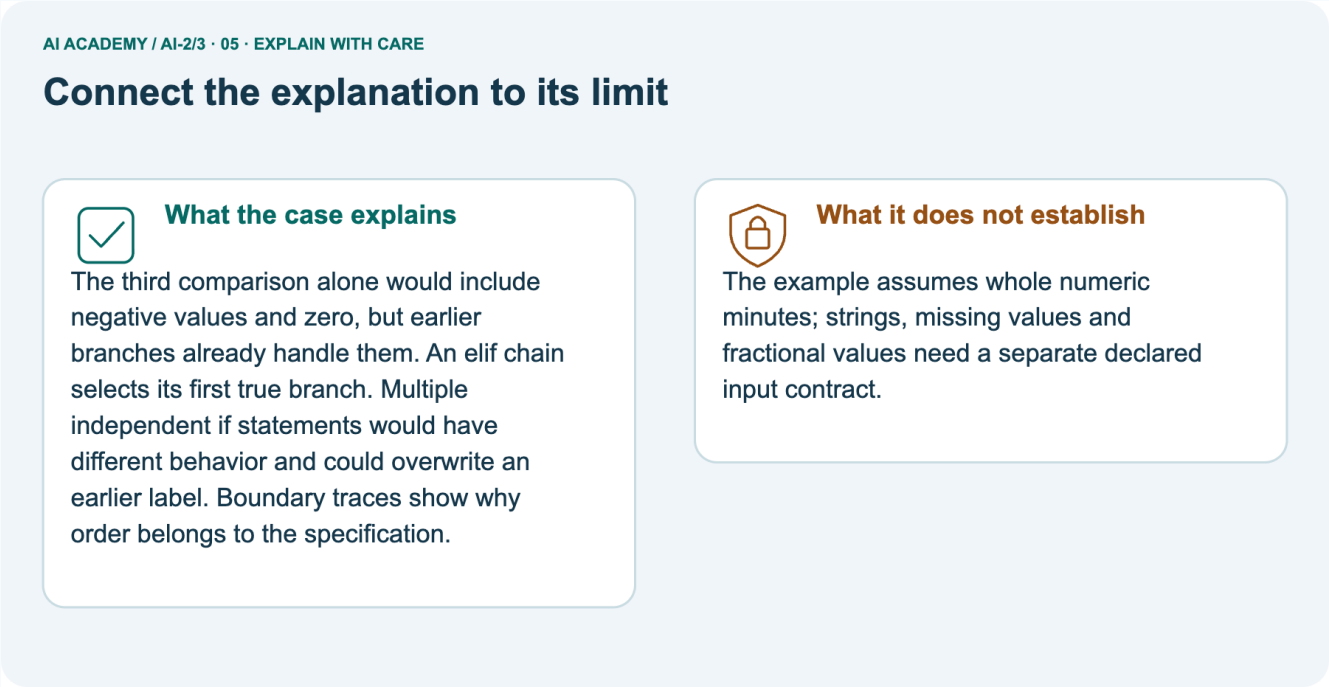
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: 11 takes Long; only the selected **branch** assigns the label. Changed **condition**: Move minutes<=10 to the first **branch**. Expected result: −1 and 0 wrongly receive Short.

Reason: Both satisfy the earlier comparison before their intended checks are reached.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

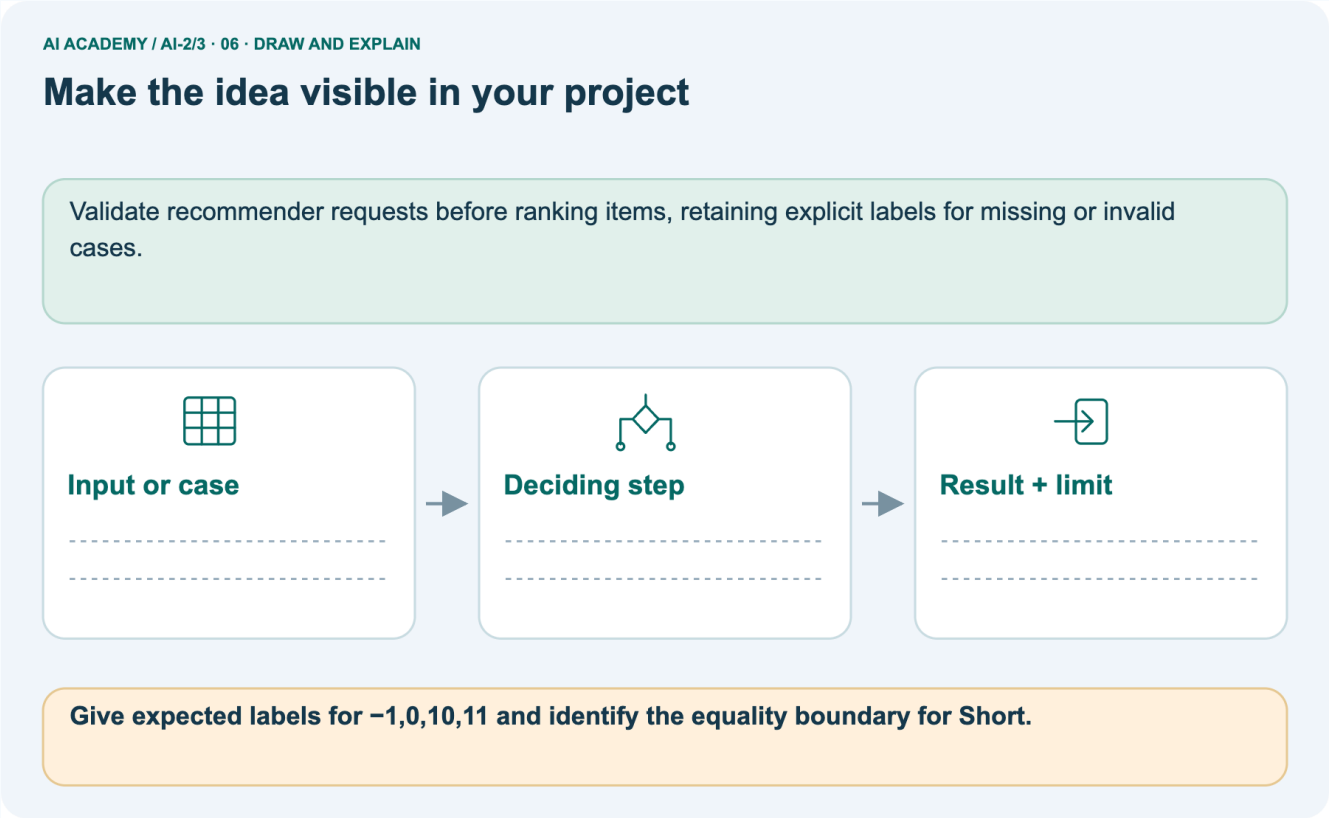
The third comparison alone would include negative values and zero, but earlier **branch**es already handle them. An elif chain selects its first true **branch**.

Multiple independent if statements would have different behavior and could overwrite an earlier label. **Boundary** traces show why **order** belongs to the specification.

Limit: The example assumes whole numeric minutes; strings, missing values and fractional values need a separate declared **input** contract.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Validate recommender requests before ranking items, retaining explicit labels for missing or invalid cases. Draw your **input**, deciding step, result and limit.

Supported question: Give expected labels for −1,0,10,11 and identify the equality **boundary** for Short.

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Branching Logic: the case

02. Follow the deciding steps

03. One chain chooses one label

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.